

Paper 5.75 - Oloid Path Carrier as Next-State Precondition

CQE-paper-05.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-05.75.md

Paper 5.75 - Oloid Path Carrier as Next-State Precondition

Purpose

Paper 5.75 explains how the rolling path carrier becomes a precondition for Paper 6 and later accumulated-center papers.

Exported State

Paper 5 exports:

```
rolling trace
legal adjacent-step relation
head/tail dyads
payload attachment rule
invalid-bit rejection
discontinuous-jump rejection
scope boundary: no prediction claim
```

Use in Paper 6

Paper 6 may register Paper 5 outputs as causal proof edges. The edge must preserve:

```
source paper
target paper
edge type
receipt
status
```

Paper 5 proves path continuity. Paper 6 proves that the dependency graph records that continuity without hiding obligations.

Use in Paper 7

Paper 7 may sample or visualize a carried trace only if it preserves the original discrete receipt. Interpolation or presentation cannot erase the rolling trace that Paper 5 proves.

Use in Later Papers

Later papers may use Paper 5 as the carrier for:

- accumulated center state,
- Hamiltonian window inputs,
- mass-residue carriers,
- lattice-boundary payload transport,
- product and adapter workflows.

Every use must state whether it imports the finite carrier theorem or proves a new prediction/physics claim.

Precondition Rule

Before a later paper uses Paper 5, it should be able to answer:

```
what is the input stream?
what is the current rolling state?
```

Which adjacent step proves continuity?
What payload is carried?
Does the payload alter the path rule?
Which receipt proves the carrier fact?

Conclusion

Paper 5.75 turns the rolling path theorem into portable state. It exports a legal carrier that can move repaired constraints forward while keeping prediction and physical interpretation tied to their own later receipts.